

Regime-Conditioned Data-Driven Subgrid-Scale Closures with Uncertainty for Turbulent Flows Past Idealized Capes and Islands

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ABSTRACT

Subgrid-scale (SGS) closures for LES of turbulent flows past obstacles must represent unresolved dynamics while remaining accurate under nonstationary forcing and stable in a posteriori integration. Most machine-learning closures, however, assume stationary statistics and provide deterministic estimates. We develop a data-driven, probabilistic two-stage convolutional neural network-sparse variational Gaussian process (CNN-SVGP) SGS closure model for two-dimensional turbulent flows. We illustrate results for nonstationary flows past idealized islands and coastal capes. In the first stage, a compact CNN maps filtered fields (e.g., vorticity, velocity, stream function, and obstacle distance) to the SGS closure terms, conditioned on parameters of the nonstationary forcing (e.g., instantaneous inflow nondimensional numbers and their recent history). With this conditioning, our model can distinguish between flows with similar instantaneous forcing but different regime-change histories. Inspired by classical wall-modeled LES, the closure target is normalized to account for near-obstacle flow inhomogeneities. Additionally, a loss term constraining the global SGS enstrophy transfer ensures physical consistency and stability in a posteriori evaluation. In the second stage, an SVGP is trained to capture the residual error of the deterministic CNN and its pointwise uncertainty. For training, the closure models employ an ensemble of simulations spanning distinct nonstationary forcing regimes. We evaluate the closures a priori and a posteriori, assessing uncertainty and confidence estimation, and generalization to unseen forcing regimes, nondimensional numbers, and obstacle geometries. The CNN-SVGP closure recovers the SGS terms accurately a priori and remains stable a posteriori. We find that prediction errors and uncertainty increase together for unseen forcing regimes and unseen nondimensional numbers, indicating that the model can accurately assess its predictive confidence.